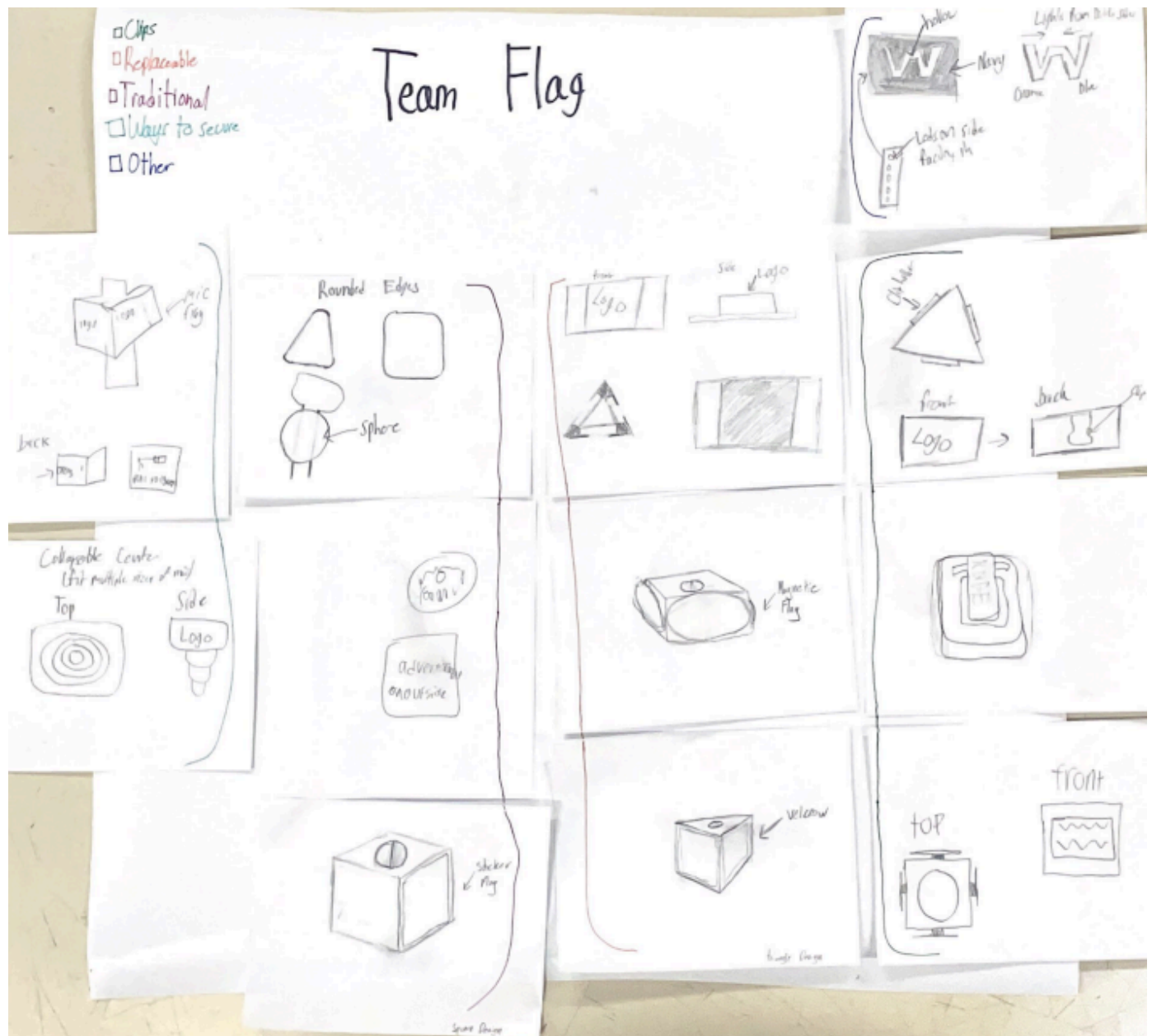


Element D

Element D: Design Concept Generation, Analysis, and Selection

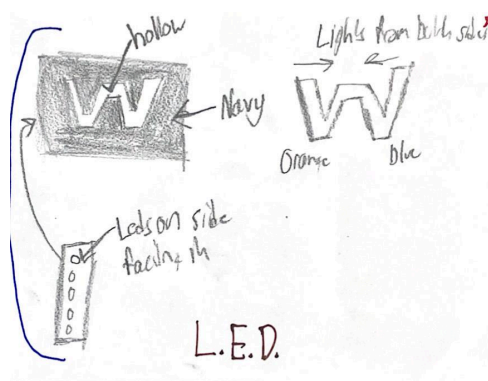
Design Idea Generation (D1)



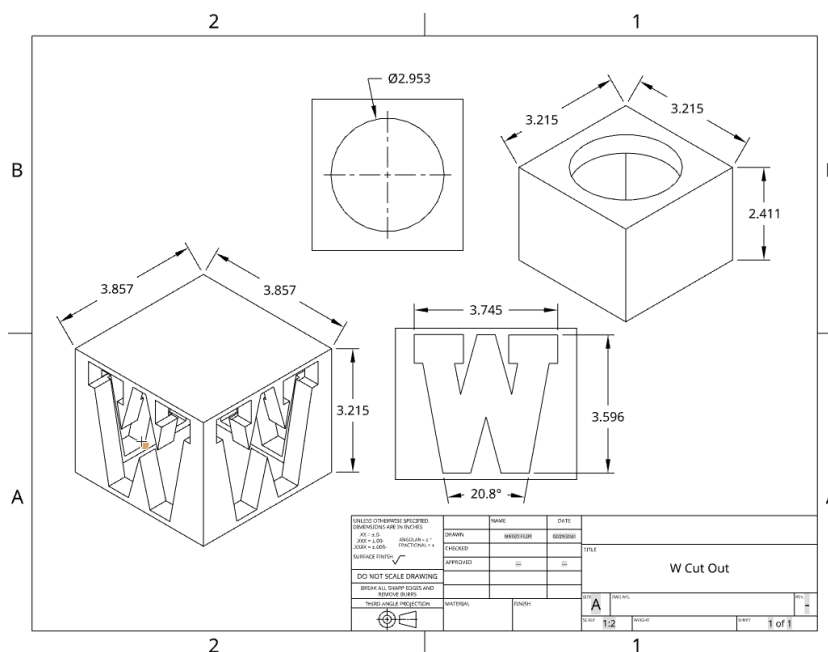
Design Idea #1 - LED Flag

Date: 2/15/24

Description: This flag will have a cut out of a W on it which will be lined with LEDs on the inside. The LEDs will light up illuminating the logo. When drawing, this idea transformed from LEDs to a W cut out with a different solid color underneath as Mr O'Farrell thought that would be a better idea.



Rough Sketch

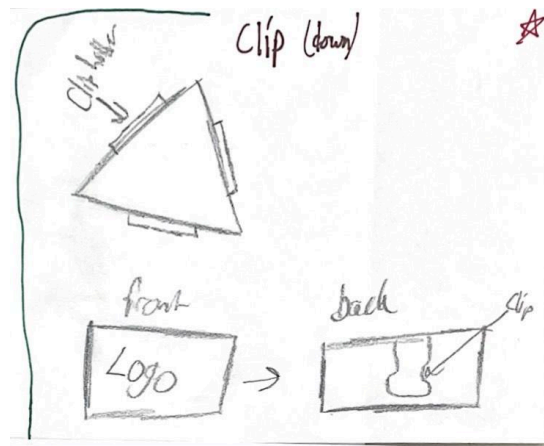


CAD Drawing

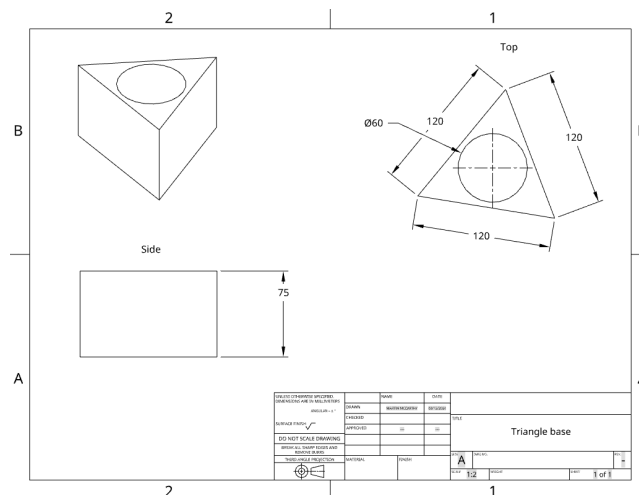
Design Idea #2 - Triangle With Clips

Date: 2/15/24

Description: Triangular shaped mic flag that uses clips to secure advertising to microphone. Drawing that includes distinguishing information such as scale, dimensions, and materials: uses 3d printing and clips. The logos will be printed on printer paper and cardstock as we will decide during testing which design makes more sense. We will tape our paper logo to the flag creating a laminate. .



Rough Sketch

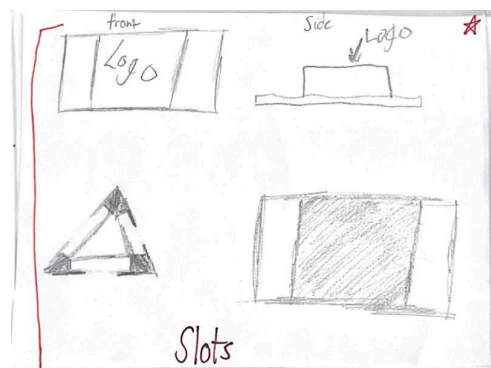


CAD Drawing

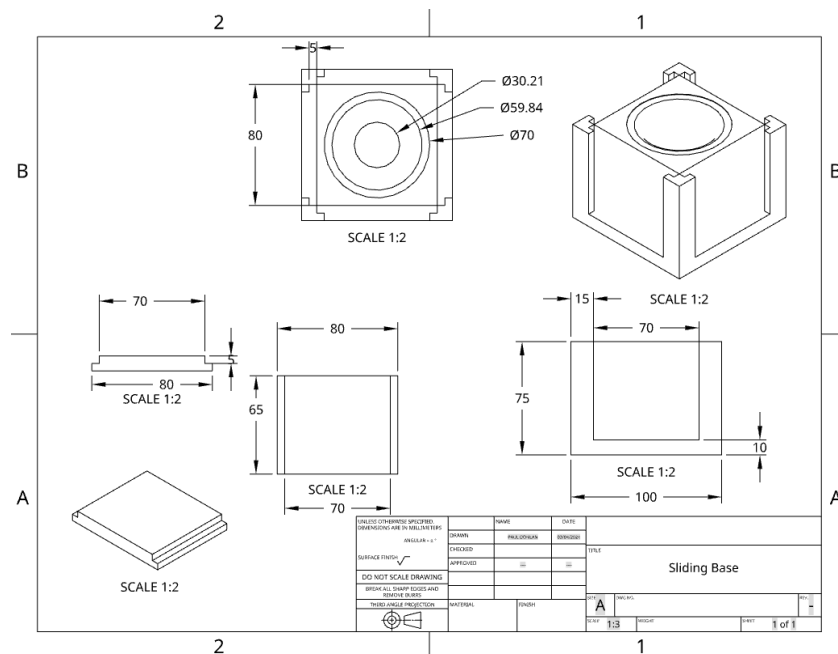
Design Idea #3 - Mic Flag With Slides

Date: 2/15/24

Description: This flag has slots for sliding logos to fit in, we are using the 3d printer and printing in color on printer paper/cardstock. The logos will be attached by tape to the slides. We chose tape because the tape will act as a laminate protecting the logo from wear and tear. The idea of the slides will make the flag easily interchangeable as logos can be slid into the flag and taken off the flag easily.



Rough Sketch



CAD Drawing

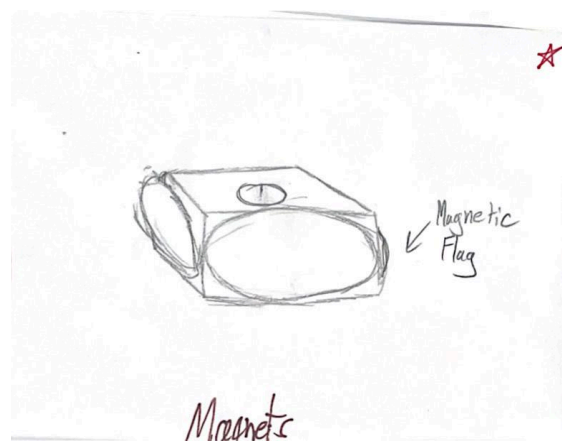
Design Idea #4 - Magnetic Flag

Date: 2/15/24

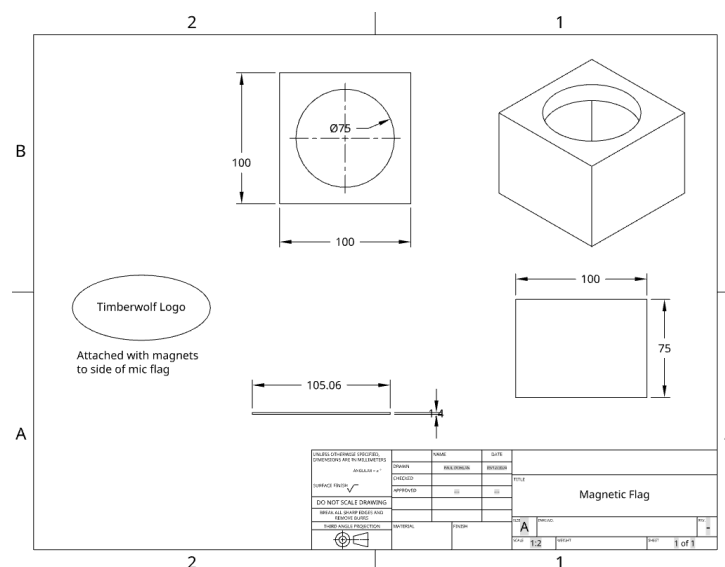
Description: Mic flag that uses magnets to secure advertising and secure to microphone

Drawing that includes distinguishing information such as scale, dimensions, and materials:

3d printing and magnets. This design will use the same logos as the sliding flag as we will print the logos on printer paper/cardstock and tape them to the magnetic panels using clear tape.



Rough Sketch



CAD Drawing

Design Idea Comparison and Selection (D1, D2)

	weight	LED	Clip	Slide	Magnets
Presentability	x2	3	4	5	4
Reproducibility	x2	1	4	4	3
Durability	x1	1	2	3	4
Price	x1	1	5	5	3
Replaceability	x1	1	4	5	4
score		1.57	3.86	4.43	3.57

Scoring: (5 Best, 1 Worst)

Presentability - How The Flag Will Look On Camera

Reproducibility - How Easy The Flag Is To Reproduce

Durability - How Strong Is The Flag When Dropped/Hit

Price - How Much Money To Make The Flag

Replaceability - How Easy It Is To Change What Logos Are On The Flag

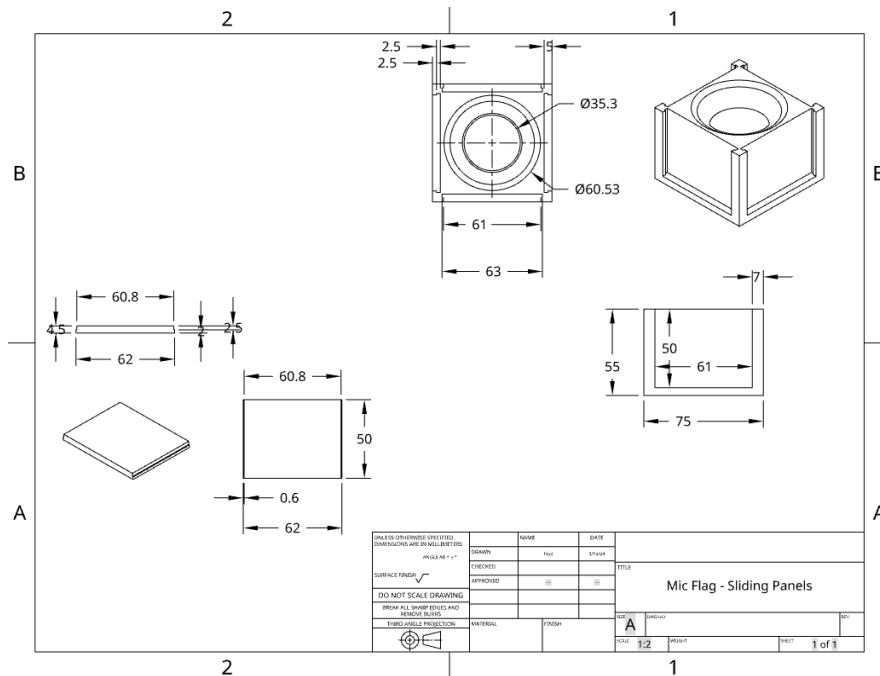
We have chosen to go with the slide mic flag as it scored the best on our matrix. We will also do a modified version of the LED with an orange inside and a blue cutout to go over the top as Mr O'Farrell thought that the mic flag with two colors and the W cut out would look really cool.

Design Blueprint (D4)

Design Blueprint #1

Date: 3/14/24

Sketch: Mic Flag With Slides



Manufacturing Methods: Onshape & 3D Printer

Step 1: Follow the sketch to make the design on onshape

Step 2: Export the design as STL file

Step 3: Slice design in Prusa (0.2 mm Speed, 15% Infill, Generic PLA), export as G Code and send it to the 3D printer

Step 4: Print the mic flag

Step 5: Glue foam inside the slot for the mic holder so that the flag will not scrape the holder.

Step 6: Print logo using color printer

Step 7: Tape over logo to prevent wear

Supply List and Costs:

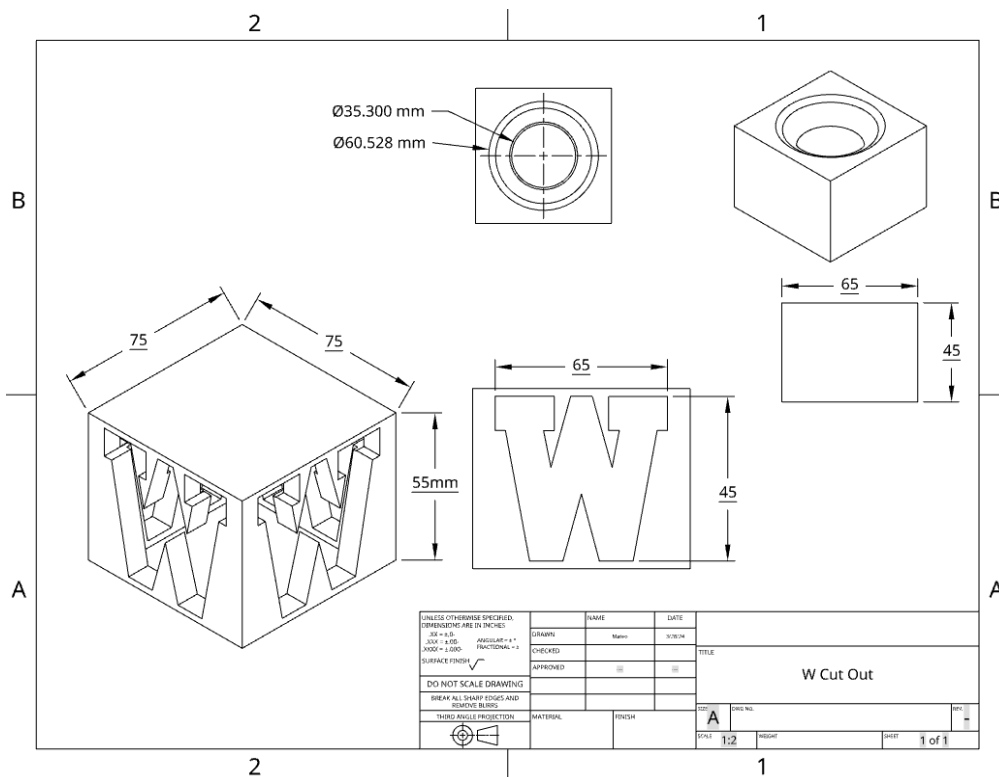
Item	Cost each	# needed	Total
1 KG 3D Printer PLA Filament	\$25	107 G	\$2.68
Soft foam	\$25	7504 mm ² (of 1,114,836 mm ²)	\$0.16
Paper	\$10	1 Sheet (of 500)	\$0.02
Clear Tape	\$3.60	4 Pieces of Tape	\$0.10
Printer Ink	\$55	4 50mmX60mm Rectangles	\$0.20
Total Cost			\$3.16



Design Blueprint #2

Date: 3/28/24

Sketch: Mic Flag With W Cut Out



Manufacturing Methods: Onshape & 3D Printer

Step 1: Follow the sketch to make the design on onshape

Step 2: Export the inside and outside piece separately as STL file

Step 3: Slice each part in Prusa (0.2 mm Speed, 15% Infill, Generic PLA), export as G Code and send it to the 3D printer

Step 4: Print the two pieces

Step 5: Assemble the flag by sliding the outside piece over the inside piece

Step 6: Glue foam inside the slot for the mic holder so that the flag will not scrape the holder.

Supply List and Costs:

Item	Cost each	# needed	Total
1 KG 3D Printer PLA Filament	\$25	150 G	\$3.75
Soft foam	\$25	7504 mm ² (of 1,114,836 mm ²)	\$0.16
Total Cost			\$3.91

Plan of Action (D3)

1. Sketch Designs **(1-2 Classes)**
2. Make Design In Onshape & Export as STL **(1-2 Classes)**
3. Slice Flag in Prusa & Print The First Mic Flag **(1 Class)**
 - 0.2 mm Speed, 15% Infill, Generic PLA
4. Test The Mic Flag **(2-3 Classes)**
 - *Testing Requirements From Element C*
 - Test Size
 - Test Fit Of Pieces
 - Test Fading Of Logos
 - Test Durability
 - Test Fit On Multiple Mic Holders Of Different Sizes
 - Test Sharpness Of Corners
 - Test Weight Of Flag
 - Assess Ease Of Replaceability
5. Record Strengths/Weaknesses **(1 Class)**
6. Make Innovations In Prototypes & Onshape **(2-3 Classes)**
7. Print New Mic Flag With New Innovations **(1 Class)**
8. Repeat Steps 2-5 Until Mic Flag Meets Our Design Requirements **(6-8 Classes To Repeat Steps 2-5)**
9. Produce Multiple Mic Flags With Different Logos/Shapes Based On Final Design After Going Through Steps 1-6 Until We Our Satisfied With Final Product **(1 Class For Each New Mic Flag To Be Manufactured)**
10. After you are satisfied with the first design (Sliding Panels), repeat steps 1 to 9 with the second design (W Cut Out) until satisfied with that design as well.